Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

An electromagnetic wave in vacuum has the electric and magnetic field $\vec{E}$ and $\vec{B}$, which are always perpendicular to each other. The direction of polarization is given by $\vec{X}$ and that of wave propagation by $\vec{k}$. Then

(A) $\vec{X} || \vec{B}$ and $\vec{k} || \vec{B} \times \vec{E}$

(B) $\vec{X} || \vec{E}$ and $\vec{k} || \vec{E} \times \vec{B}$

(C) $\vec{X} || \vec{B}$ and $\vec{k} || \vec{E} \times \vec{B}$

(D) $\vec{X} || \vec{E}$ and $\vec{k} || \vec{B} \times \vec{E}$

Q.2

If E and B represent electric and magnetic field vectors of the electromagnetic wave, the direction of propagation of electromagnetic wave is along

(A) $\overrightarrow{\mathrm{E}}$

(B) $\overline{\mathrm{B}}$

(C) $\overrightarrow{\mathrm{B}} \times \overrightarrow{\mathrm{E}}$

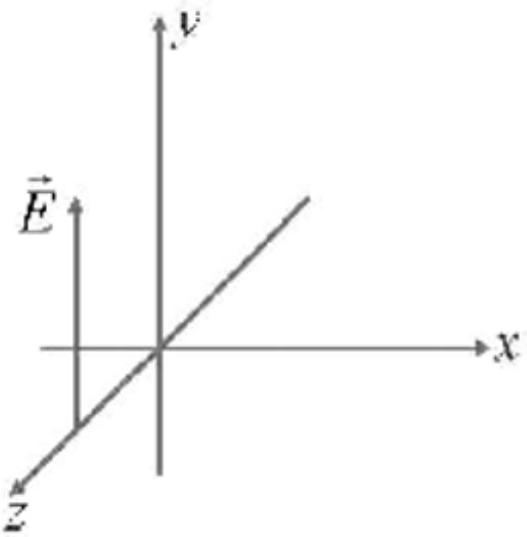
(D) $\overrightarrow{\mathrm{E}} \times \overrightarrow{\mathrm{B}}$

Q.3

The figure here gives the electric field of an electromagnetic wave at a certain point and a certain instant. The wave is transporting energy in the negative z-direction. The direction of the magnetic field of the wave at that point and

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instant is



(A) $+$ ve x -direction

(B) $-$ ve x -direction

(C) $+$ ve z -direction

(D) $-$ ve y -direction

Q.4

A plane electromagnetic wave is incident on a material surface. If the wave delivers momentum p and energy E, then

(A) $\mathrm{p}=0, \mathrm{E}=0$

(B) $\mathrm{p} \neq 0, \mathrm{E} \neq 0$

(C) $\mathrm{p} \neq 0, \mathrm{E}=0$

(D) $\mathrm{p}=0, \mathrm{E} \neq 0$

Q.5

The electric field of an electromagnetic wave travelling through vaccum is given by the equation $E = E_0 \sin(kx - \omega t)$. The quantity that is independent of wavelength is

(A) $k\omega$

(B) $\frac{k}{\omega}$

(C) $k^2\omega$

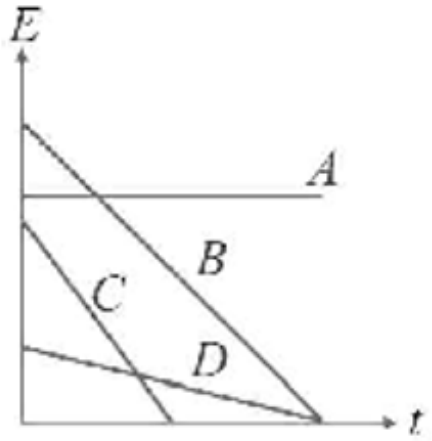
(D) ω

Q.6

The figure shows graphs of the electric field magnitude E versus time t for four uniform electric fields, all

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contained within identical circular regions. Which of them is according to the magnitudes of the magnetic field?



(A) A

(B) B

(C) C

(D) D

Q.7

The electric field part of an electromagnetic wave in a medium is represented by $\mathrm{E_x} = 0$

$$\mathrm{E_y} = 2.5\frac{\mathrm{N}}{\mathrm{C}}\cos\left[\left(2\pi \times 10^6 \frac{\mathrm{rad}}{\mathrm{m}}\right)\mathrm{t} - \left(\pi \times 10^{-2}\frac{\mathrm{rad}}{\mathrm{s}}\right)\mathrm{x}\right]$$

$\mathrm{E}_z = 0$. The wave is:

(A) moving along x direction with frequency 10^6 Hz and wave length 100 m.

(B) moving along x direction with frequency 10^6 Hz and wave length 200 m.

(C) moving along $-x$ direction with frequency 10^6 Hz and wave length 200 m.

(D) moving along y direction with frequency $2\pi \times 10^6$ Hz and wave length 200 m.

Q.8

The speed of electromagnetic wave in vacuum depends upon the source of radiation. It

(A) increases as we move from γ -rays to radio waves

(B) decreases as we move from γ -rays to radio waves

(C) is same for all of them

(D) None of these

Q.9

The energy of electromagnetic wave in vacuum is given by the relation

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(A) $\frac{E^2}{2\varepsilon_0}+\frac{B^2}{2\mu_0}$

(B) $\frac{1}{2}\varepsilon_0 \mathrm{E}^2+\frac{1}{2}\mu_0 \mathrm{B}^2$

(C) $\frac{\mathrm{E}^2+\mathrm{B}^2}{\mathrm{c}}$

(D) $\frac{1}{2}\varepsilon_0 \mathrm{E}^2+\frac{\mathrm{B}^2}{2\mu_0}$

Q.10

Out of the following options which one can be used to produce a propagating electromagnetic wave?

(A) A charge moving at constant velocity

(B) A stationary charge

(C) A chargeless particle

(D) An accelerating charge

Q.11

The pressure exerted by an electromagnetic wave of intensity I (watts $/\mathrm{m}^2$) on a non reflecting surface is $[\mathrm{c}$ is the velocity of light $]$

(A) Ic

(B) Ic^2

(C) I/c

(D) I/c^2

Q.12

Which of the following statement is false for the properties of electromagnetic waves?

(A) Both electric and magnetic field vectors attain the maxima and minima at the same place and same time.

(B) The energy in electromagnetic wave is divided equally between electric and magnetic vectors

(C) Both electric and magnetic field vectors are parallel to each other and perpendicular to the direction of propagation of wave

(D) These waves do not require any material medium for propagation.

Q.13

A radiation of energy ' E ' falls normally on a perfectly reflecting surface. The momentum transferred to the surface is $(\mathrm{c}=$ Velocity of light $)$

(A) $\frac{2\mathrm{E}}{\mathrm{c}}$

(B) $\frac{2\mathrm{E}}{\mathrm{c}^2}$

(C) $\frac{E}{c^2}$

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(D) $\frac{E}{c}$

Q.14

An electromagnetic wave going through vacuum is described by $E = E_0 \sin(kx - \omega t); B = B_0 \sin(kx - \omega t)$. Which
of the following equations is true

(A) $E_0 k = B_0 \omega$

(B) $E_0 \omega = B_0 k$

(C) $E_0\ B_0 = \omega k$

(D) None of these

Q.15

If c is the speed of electromagnetic waves in vacuum, its speed in a medium of dielectric constant K and relative permeability μ_r is

(A) $v = \frac{1}{\sqrt{\mu_r K}}$

(B) $v = c\sqrt{\mu_r K}$

(C) $v = \frac{c}{\sqrt{\mu_r K}}$

(D) $v = \frac{K}{\sqrt{\mu_r c}}$

Q.16

A point source of electromagnetic radiation has an average power output of $1500\ W$. What will be the maximum value of electric field at a distance of $3\ m$ from this sources in Vm^{-1}?

(A) 10

(B) 73

(C) 1000

(D) 500

Q.17

A plane electromagnetic wave in a non-magnetic dielectric medium is given by $\vec{E} = \bar{E}_0 \left(4 \times 10^{-7} x - 50t\right)$ with distance being in meter and time in seconds. What is value of dielectric constant of the medium?

(A) 5.7

(B) 5.8

(C) 5.9

(D) 6.0

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Q.18

A plane electromagnetic wave travels in free space along x -axis. At a particular point in space, the electric field along y-axis is $9.3\ \mathrm{V\ m^{-1}}$. Find the magnetic induction (in Tesla) along z-axis. Assuming 1 decimal place in answer expression, what is the power of 10 ?

(A) -7

(B) -8

(C) -9

(D) -10

Q.19

The rms value of the electric field of the light coming from the Sun is $720\ \mathrm{N/C}$. What is the average total energy density (in $\mathrm{J/m^3}$) of the electromagnetic wave? Assuming 2 decimal places in answer expression, what is the power of 10 ?

(A) -3

(B) -4

(C) -5

(D) -6

Q.20

The magnetic field in the plane electromagnetic field is given by:

$\mathrm{B_y} = 2 \times 10^{-7} \sin\left(0.5 \times 10^3 \mathrm{z} + 1.5 \times 10^{11} \mathrm{t}\right) \mathrm{T}$

The expression for the electric field may be given by

(A) $\mathrm{E_y} = 2 \times 10^{-7} \sin\left(0.5 \times 10^3 \mathrm{z} + 1.5 \times 10^{11} \mathrm{t}\right) \mathrm{V/m}$

(B) $\mathrm{E_x} = 2 \times 10^{-7} \sin\left(0.5 \times 10^3 \mathrm{z} + 1.5 \times 10^{11} \mathrm{t}\right) \mathrm{V/m}$

(C) $\mathrm{E_y} = 60 \sin\left(0.5 \times 10^3 \mathrm{z} + 1.5 \times 10^{11} \mathrm{t}\right) \mathrm{V/m}$

(D) $E_x = 60 \sin\left(0.5 \times 10^3 z + 1.5 \times 10^{11} \mathrm{t}\right) \mathrm{V/m}$

Answers & Solutions

Q.1 Answer:

$\vec{X} \| \vec{E}$ and $\vec{k} \| \vec{E} \times \vec{B}$

Solution:

$\because$ The E.M. wave are transverse in nature i.e.,

$= \frac{\vec{k} \times \vec{E}}{\mu\omega} = \vec{H}$

where $\vec{H} = \frac{\vec{B}}{\mu}$

and $\frac{\vec{k} \times \vec{H}}{\omega\varepsilon} = -\vec{E}$

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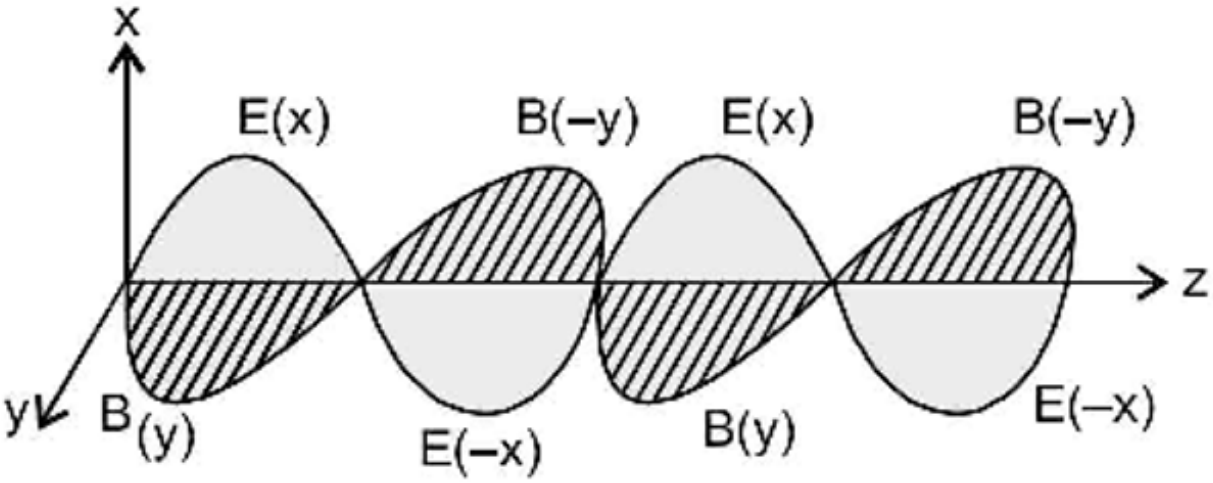
$\vec{k}$ is $\perp$ $\vec{H}$ and $\vec{k}$ is also $\perp$ to $\vec{E}$
or In other words $\vec{X} \| \vec{E}$ and $\vec{k} \| \vec{E} \times \vec{B}$

Q.2 Answer:

$\overrightarrow{\mathrm{E}} \times \overrightarrow{\mathrm{B}}$

Solution:

The direction of propagation of electromagnetic wave is perpendicular to both electric field E and magnetic field B, i.e., in the direction of $\mathrm{E} \times \mathrm{B}$ by right thumb rule. The diagram given below



So, electromagnetic wave is along the z-direction which is give the cross product of E and B direction is perpendicular to E and B from $\overrightarrow{\mathrm{E}}$ to $\overrightarrow{\mathrm{B}}$. i.e., $(\mathrm{E} \times \mathrm{B})$ in $\mathrm{z}-$ direction.

Q.3 Answer:

$+$ ve x -direction

Solution:

Direction of energy progration of EM-waves is given by
$\vec{D} = K(\vec{E} \times \vec{B})$
or $-\hat{k} = K(E\hat{j} \times \bar{B})$
Clearly direction of magnetic field is along positive x axis.

Q.4 Answer:

$\mathrm{p} \neq 0, \mathrm{E} \neq 0$

Solution:

EM waves carry momentum and hence can exert pressure on surfaces. They also transfer energy to the surface so $\mathrm{p} \neq 0$ and $\mathrm{E} \neq 0$

Q.5 Answer:

$\frac{k}{\omega}$

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Solution:

Here, $k = \frac{2\pi}{\lambda}, \omega = 2\pi \mathrm{v}$
$\therefore \quad \frac{k}{\omega} = \frac{2\pi/\lambda}{2\pi v} = \frac{1}{\pi v} = \frac{1}{c} \quad (\because c = v\lambda)$
where c **is the speed of electromagnetic wave in vacuum. It is a constant whose value is** $3 \times 10^6 \ \mathrm{ms}^{-1}$

Q.6 Answer:

C

Solution:

$\oint \vec{B} \cdot d\bar{\ell} = \mu_0 \in_0 \frac{d\phi}{dt}$
or $B \times 2\pi r = \mu_0 \epsilon_0 A \left(\frac{dE}{dt}\right) \quad \therefore B \propto \left(\frac{dE}{dt}\right)$

Q.7 Answer:

moving along x direction with frequency $10^6 \ \mathrm{Hz}$ and wave length $200 \ \mathrm{m}$.

Solution:

Comparing with the equation of wave. $E_y = E_0 \cos(\omega t - kx)$
$\omega = 2\pi \mathrm{f} = 2\pi \times 10^6 \quad \therefore \mathrm{f} = 10^6 \ \mathrm{Hz}$
$\frac{2\pi}{\lambda} = \mathrm{k} = \pi \times 10^{-2} \ \mathrm{m}^{-1}, \lambda = 200 \ \mathrm{m}$

Q.8 Answer:

is same for all of them

Solution:

Speed of EM waves in vacuum $= \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ **= constant**

Q.9 Answer:

$\frac{1}{2}\varepsilon_0 \mathrm{E}^2 + \frac{\mathrm{B}^2}{2\mu_0}$

Solution:

$\frac{1}{2}\varepsilon_0 \mathrm{E}_0^2$ **is electric energy density.**
$\frac{\mathrm{B}^2}{2\mu_0}$ **is magnetic energy density.**
So, total energy $= \frac{1}{2}\varepsilon_0 \mathrm{E}_0^2 + \frac{\mathrm{B}_0^2}{2\mu_0}$

Q.10 Answer:

An accelerating charge

Solution:

To generate electromagnetic waves we need accelerating charge particle.

Q.11 Answer:

$\mathrm{I/c}$

Solution:

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Pressure $= I/c$

Q.12 Answer:

Both electric and magnetic field vectors are parallel to each other and perpendicular to the direction of propagation of wave

Solution:

Electromagnetic waves are the combination of mutually perpendicular electric and magnetic fields.

Q.13 Answer:

$\frac{2E}{c}$

Solution:

Momentum of light falling on reflecting surface $p = \frac{E}{c}$ **As surface is perfectly reflecting so momentum reflect**

$p^1 = -\frac{E}{c}$



So, momentum transferred

$= P - P^1 = \frac{E}{c} - \left(-\frac{E}{c}\right) = \frac{2E}{c}$

Q.14 Answer:

$E_0 k = B_0 \omega$

Solution:

$\frac{E_0}{B_0} = c$. **also** $k = \frac{2\pi}{\lambda}$ **and** $\omega = 2\pi V$

These relation gives $E_0 k = B_0 \omega$

Q.15 Answer:

$v = \frac{c}{\sqrt{\mu_r K}}$

Solution:

Speed of light of vacuum $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ **and in another medium** $v = \frac{1}{\sqrt{\mu \varepsilon}}$

$\therefore \frac{c}{v} = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}} = \sqrt{\mu_r K} \Rightarrow v = \frac{c}{\sqrt{\mu_r K}}$

Q.16 Answer:

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73

Solution:

Intensity of EM wave is given by:-

$$\mathrm{I} = \frac{\mathrm{P}}{4\pi \mathrm{R}^2}$$
$$= \mathrm{V}_{\mathrm{av}} \cdot \mathrm{c} = \frac{1}{2}\varepsilon_0 \mathrm{E}_0^2 \times \mathrm{c}$$
$$\Rightarrow \mathrm{E}_0 = \sqrt{\frac{\mathrm{P}}{2\pi \mathrm{R}^2 \varepsilon_0 \mathrm{c}}}$$
$$= \sqrt{\frac{1500}{2\times 3.14\times (3)^2 \times 8.85\times 10^{-12}\times 3\times 10^8}}$$
$$= 73 \frac{\mathrm{V}}{\mathrm{m}}$$

Q.17 Answer:

5.8

Solution:

$$\vec{E} = \vec{E}_0(4 \times 10\mathrm{x} - 50t)$$
$$\vec{E} = \vec{E}_0(kx - wt)$$

we get,

$$\mathrm{w} = 50 \mathrm{rad/s.k} = 4 \times 10^{-7}\ \mathrm{m}^{-1}$$

Thus velocity of wave,

$$\mathrm{v} = \frac{\mathrm{w}}{\mathrm{k}} = \frac{50}{4\times 10^{-7}} = 1.25 \times 10^8\ \mathrm{m/s}$$

so, refractive index of medium,

$$\mu = \frac{\mathrm{e}}{\mathrm{v}} = \frac{3\times 10^8}{1.25\times 10^8} = 2.4$$

using,

$$\mathrm{u}^2 = \mathrm{k}$$

as medium is non diamagnetic, $\mathrm{km} = 1$

$$\Rightarrow \mathrm{ke} = \mu^2 = (2.4)^2 \Rightarrow \mathrm{ke} = 5.76$$

Q.18 Answer:

-8

Solution:

$$\left(3.1 \times 10^{-8}\right)$$

Velocity of light

$$\mathrm{C} = \frac{\mathrm{E}}{\mathrm{B}} \Rightarrow \mathrm{B} = \frac{\mathrm{E}}{\mathrm{C}} = \frac{9.3}{3\times 10^8} = 3.1 \times 10^{-8}\ \mathrm{T}$$

Q.19 Answer:

-6

Solution:

$$\left(4.58 \times 10^{-6}\right)$$
$$E_{\mathrm{ms}} = 720$$

The average total energy density

$$= \frac{1}{2} \in_0 \mathrm{E}_0^2 = \frac{1}{2}\epsilon_0 \left[\sqrt{2}\mathrm{E}_{\mathrm{rms}}\right]^2 = \epsilon_0 \mathrm{E}_{\mathrm{rms}}^2$$

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$= 8.85 \times 10^{-12} \times (720)^2$

$= 4.58 \times 10^{-6}\ \mathrm{J/m^3}$

Q.20 Answer:

$E_x = 60 \sin\left(0.5 \times 10^3 z + 1.5 \times 10^{11} \mathrm{t}\right) \mathrm{V/m}$

Solution:

$\mathrm{B_y} = 2 \times 10^{-7} \sin\left(0.5 \times 10^3 \mathrm{z} + 1.5 \times 10^{11} \mathrm{t}\right) \mathrm{T}$

The electric vector is perpendicular to B as well as direction of propagation of electromagnetic wave. Therefore $\mathrm{E_x}$ has to be taken. Further, $\mathrm{E_0} = \mathrm{B_0} \times \mathrm{c}$

$= 2 \times 10^{-7} \times 3 \times 10^8\ \mathrm{V/m}$

$\mathrm{E_0} = 2 \times 10^{-7} \times 3 \times 10^8 = 60\ \mathrm{V/m}$

$\therefore$ **The corresponding value of the electric field is**

$\mathrm{Ex} = 60 \sin\left(0.5 \times 10^3 \mathrm{z} + 1.5 \times 10^{11} \mathrm{t}\right) \mathrm{V/m}$

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